Differential Geometry

Exercise 13

Ari Feiglin

13.1 Problem (DoCarmo, Riemannian Geometry, page 106 question 8)

Let M be a connected Riemannian manifold of dimension $n \geq 3$. Suppose that M is **isotropic**: its sectional curvature at each point is independent on choice of subspace $\sigma \subseteq T_pM$. Prove that M has constant sectional curvature; that is its sectional curvature is further independent on choice of point.

Let us define the 4-tensor

$$\mathrm{Rm}'(W, Z, X, Y) = \langle W, X\rangle\langle Z, Y\rangle - \langle Z, X\rangle\langle W, Y\rangle.$$

Because sectional curvature $K(p, \sigma) = K(p)$ is independent of σ, we know that $\mathrm{Rm} = K\mathrm{Rm}'$. Notice that for $U \in \mathfrak{X}(M)$,

$$\nabla_U \mathrm{Rm}'(W, Z, X, Y) = U(\mathrm{Rm}'(W, Z, X, Y)) - \langle\nabla_U W, X\rangle\langle Z, Y\rangle + \langle Z, X\rangle\langle\nabla_U W, Y\rangle - \cdots.$$

Because the connection is metric, further expansions of the terms show that this all cancels out. This is lengthy so I will not write everything out, but for example expanding once gives

$$U(\mathrm{Rm}'(W, Z, X, Y)) = U\langle W, X\rangle\langle Z, Y\rangle - U\langle Z, X\rangle\langle W, Y\rangle = \langle\nabla_U W, Z\rangle\langle Z, Y\rangle - \langle Z, X\rangle\langle\nabla_U W, Y\rangle + \cdots$$

which cancels out with the first term above.

Thus

$$\nabla_U \mathrm{Rm}' = 0$$

and so

$$\nabla\mathrm{Rm}(\bullet, \bullet, \bullet, \bullet, U) = \nabla_U \mathrm{Rm} = \nabla_U(K\mathrm{Rm}') = (UK)\mathrm{Rm}' + K\nabla_U\mathrm{Rm}' = (UK)\mathrm{Rm}'.$$

The second Binachi identity gives

$$\nabla\mathrm{Rm}(W, Z, X, Y, U) + \nabla\mathrm{Rm}(W, Z, Y, U, X) + \nabla\mathrm{Rm}(W, Z, U, X, Y) = 0.$$

Thus

$$(UK)\mathrm{Rm}'(W, Z, X, Y) + (XK)\mathrm{Rm}'(W, Z, Y, U) + (YK)\mathrm{Rm}'(W, Z, U, X) = 0.$$

Fix a $p \in M$ and let $X \in T_pM$ be nonzero. Because $n \geq 3$ we can find $Y, Z \in T_pM$ such that $\langle X, Y\rangle = \langle Y, Z\rangle = \langle Z, X\rangle = 0$ and $\langle Z, Z\rangle = 1$ (extend X to an orthogonal basis). Letting $U = Z$, and letting W be arbitrary we get

$$(ZK)\mathrm{Rm}'(W, Z, X, Y) + (XK)\mathrm{Rm}'(W, Z, Y, Z) + (YK)\mathrm{Rm}'(W, Z, Z, X) = 0.$$

Expanding definitions,

$$\begin{aligned}
\mathrm{Rm}'(W, Z, X, Y) &= \langle W, X\rangle\langle Z, Y\rangle - \langle Z, X\rangle\langle W, Y\rangle = 0,\\
\mathrm{Rm}'(W, Z, Y, Z) &= \langle W, Y\rangle\langle Z, Z\rangle - \langle Z, Y\rangle\langle W, Z\rangle = \langle W, Y\rangle,\\
\mathrm{Rm}'(W, Z, Z, X) &= \langle W, Z\rangle\langle Z, X\rangle - \langle Z, Z\rangle\langle W, X\rangle = -\langle W, X\rangle.
\end{aligned}$$

Thus we see that

$$(XK)\langle W, Y\rangle - (YK)\langle W, X\rangle = \langle W, (XK)Y - (YK)X\rangle = 0.$$

Since this is true for arbitrary W, this means that $(XK)Y - (YK)X = 0$. Because X, Y are linearly independent this implies $XK = 0$.

So for all $X \in T_pM$ we have $XK = 0$ (for $X = 0$ this is trivial). Thus K is locally constant, and since M is connected this means K is constant as desired.

13.2 Problem (DoCarmo, Riemannian Geometry, page 187 question 10)

Let $f\colon \overline{M} \to M$ be a Riemannian submersion; $X, Y, Z, W \in \mathfrak{X}(M)$ and $\overline{X}, \overline{Y}, \overline{Z}, \overline{W}$ be their horizontal lifts. Let $R, \overline{R}$ be the curvature tensors of $M, \overline{M}$ respectively. Prove the following.

(1)

$$\left\langle \overline{R}(\overline{X}, \overline{Y})\overline{Z}, \overline{W} \right\rangle = \langle R(X,Y)Z, W\rangle - \frac{1}{4}\left\langle [\overline{X}, \overline{Z}]^v, [\overline{Y}, \overline{W}]^v \right\rangle + \frac{1}{4}\left\langle [\overline{Y}, \overline{Z}]^v, [\overline{X}, \overline{W}]^v \right\rangle - \frac{1}{2}\left\langle [\overline{Z}, \overline{W}]^v, [\overline{X}, \overline{Y}]^v \right\rangle.$$

(2) Let σ be the plane generated by orthonormal $X, Y \in \mathfrak{X}(M)$, and $\overline{\sigma}$ the plane generated by $\overline{X}, \overline{Y}$. Then

$$K(\sigma) = \overline{K}(\overline{\sigma}) + \frac{3}{4}\left|[\overline{X}, \overline{Y}]^v\right|^2 \geq \overline{K}(\overline{\sigma}).$$

(1) Notice that, as per a previous exercise,

$$\overline{X}\left\langle \overline{\nabla}_{\overline{Y}}\overline{Z}, \overline{W} \right\rangle = \overline{X}\left\langle \overline{\nabla_X Y} + \frac{1}{2}[\overline{X}, \overline{Y}]^v, \overline{W} \right\rangle = \overline{X}\left\langle \overline{\nabla_X Y}, \overline{W} \right\rangle = X\langle \nabla_X Y, W\rangle$$

where the second equality is because $\overline{W}$ is horizontal and thus orthogonal to vertical vectors.

Because the connection is metric,

$$\left\langle \overline{\nabla}_{\overline{X}}\overline{\nabla}_{\overline{Y}}\overline{Z}, \overline{W} \right\rangle = \overline{X}\left\langle \overline{\nabla}_{\overline{Y}}\overline{Z}, \overline{W} \right\rangle - \left\langle \overline{\nabla}_{\overline{Y}}\overline{Z}, \overline{\nabla}_{\overline{X}}\overline{W} \right\rangle.$$

The first term is equal to $X\left\langle \overline{\nabla}_Y Z, W \right\rangle$. The second term is equal to

$$\begin{aligned} \left\langle \overline{\nabla}_{\overline{Y}}\overline{Z}, \overline{\nabla}_{\overline{X}}\overline{W} \right\rangle = \left\langle \overline{\nabla_Y Z} + \frac{1}{2}[\overline{Y}, \overline{Z}], \overline{\nabla_X W} + \frac{1}{2}[\overline{X}, \overline{W}]^v \right\rangle &= \left\langle \overline{\nabla_Y Z}, \overline{\nabla_X W} \right\rangle + \frac{1}{4}\left\langle [\overline{Y}, \overline{Z}]^v, [\overline{X}, \overline{W}]^v \right\rangle \\ &= \langle \nabla_Y Z, \nabla_X W\rangle + \frac{1}{4}\left\langle [\overline{Y}, \overline{Z}]^v, [\overline{X}, \overline{W}]^v \right\rangle. \end{aligned}$$

Thus

$$\begin{aligned} \left\langle \overline{\nabla}_{\overline{X}}\overline{\nabla}_{\overline{Y}}\overline{Z}, \overline{W} \right\rangle = X\langle \nabla_Y Z, W\rangle - \langle \nabla_Y Z, \nabla_X W\rangle - \frac{1}{4}\left\langle [\overline{Y}, \overline{Z}]^v, [\overline{X}, \overline{W}]^v \right\rangle \\ = \langle \nabla_X \nabla_Y Z, W\rangle - \frac{1}{4}\left\langle [\overline{Y}, \overline{Z}]^v, [\overline{X}, \overline{W}]^v \right\rangle. \end{aligned}$$

If $T \in \mathfrak{X}(\overline{M})$ is vertical, then

$$\left\langle \overline{\nabla}_T \overline{X}, \overline{Y} \right\rangle = \left\langle \overline{\nabla}_{\overline{X}} T, \overline{Y} \right\rangle + \left\langle [T, \overline{X}], \overline{Y} \right\rangle.$$

The second term is zero, as per the previous exercise. Thus this is equal to (since $\left\langle T, \overline{Y} \right\rangle = 0$)

$$= \overline{X}\left\langle T, \overline{Y} \right\rangle - \left\langle T, \overline{\nabla}_{\overline{X}}\overline{Y} \right\rangle = -\left\langle T, \overline{\nabla}_{\overline{X}}\overline{Y} \right\rangle = -\frac{1}{2}\left\langle T, [\overline{X}, \overline{Y}]^v \right\rangle.$$

Now we have

$$\left\langle \overline{\nabla}_{[\overline{X}, \overline{Y}]}\overline{Z}, \overline{W} \right\rangle = \left\langle \overline{\nabla}_{[\overline{X}, \overline{Y}]^h}\overline{Z}, \overline{W} \right\rangle + \left\langle \overline{\nabla}_{[\overline{X}, \overline{Y}]^v}\overline{Z}, \overline{W} \right\rangle.$$

As we just showed, the second term is equal to

$$\left\langle \overline{\nabla}_{[\overline{X},\overline{Y}]^v} \overline{Z}, \overline{W} \right\rangle = -\frac{1}{2} \langle [\overline{X},\overline{Y}]^v, [\overline{Z},\overline{W}]^v \rangle.$$

As per the previous homework $\langle [\overline{X},\overline{Y}]^h, \overline{Z} \rangle = \langle [\overline{X},\overline{Y}], \overline{Z} \rangle = \langle [X,Y], Z \rangle$. Thus

$$\langle [\overline{X},\overline{Y}]^h, \overline{Z} \rangle = \left\langle \overline{[X,Y]}, \overline{Z} \right\rangle = \langle [X,Y], Z \rangle$$

and since all horizontal vectors are the lift of some vector field this means that $[\overline{X},\overline{Y}]^h = \overline{[X,Y]}$. Thus

$$\left\langle \overline{\nabla}_{[\overline{X},\overline{Y}]} \overline{Z}, \overline{W} \right\rangle = \langle \nabla_{[X,Y]} Z, W \rangle - \frac{1}{2} \langle [\overline{X},\overline{Y}]^v, [\overline{Z},\overline{W}]^v \rangle.$$

Putting this together,

$$\begin{aligned}
\langle \overline{R}(\overline{X},\overline{Y})\overline{Z},\overline{W} \rangle &= \langle \overline{\nabla}_{\overline{Y}} \overline{\nabla}_{\overline{X}} \overline{Z}, \overline{W} \rangle - \langle \overline{\nabla}_{\overline{X}} \overline{\nabla}_{\overline{Y}} \overline{Z}, \overline{W} \rangle + \left\langle \overline{\nabla}_{[\overline{X},\overline{Y}]} \overline{Z}, \overline{W} \right\rangle \\
&= \langle R(X,Y)Z, W \rangle - \frac{1}{4} \langle [\overline{X},\overline{Z}]^v, [\overline{Y},\overline{W}]^v \rangle \\
&\quad + \frac{1}{4} \langle [\overline{Y},\overline{Z}]^v, [\overline{X},\overline{W}]^v \rangle - \frac{1}{2} \langle [\overline{Z},\overline{W}]^v, [\overline{X},\overline{Y}]^v \rangle.
\end{aligned}$$

(2) Because df preserves the metric of horizontal vectors, $\overline{X}, \overline{Y}$ remain orthonormal. Thus

$$\overline{K}(\overline{\sigma}) = \overline{\mathrm{Rm}}(\overline{X},\overline{Y},\overline{X},\overline{Y}) = \mathrm{Rm}(X,Y,X,Y) - \frac{1}{4} \langle [\overline{X},\overline{X}]^v, [\overline{Y},\overline{Y}]^v \rangle + \frac{1}{4} \langle [\overline{Y},\overline{X}]^v, [\overline{X},\overline{Y}]^v \rangle - \frac{1}{2} \langle [\overline{X},\overline{Y}]^v, [\overline{X},\overline{Y}]^v \rangle,$$

and because the Lie bracket is antisymmetric we get that the first inner product is zero, the second is equal to $-\left|[\overline{X},\overline{Y}]^v\right|^2/4$ and so

$$K(\sigma) = \mathrm{Rm}(X,Y,X,Y) = \overline{K}(\overline{\sigma}) + \frac{3}{4} \left|[\overline{X},\overline{Y}]^v\right|^2.$$

13.3 Problem (DoCarmo, Riemannian Geometry, page 188 question 12)

Define a Riemannian metric on $\mathbb{C}^{n+1} - 0$ as follows. For $x \in \mathbb{C}^{n+1} - 0$ and $V, W \in T_z(\mathbb{C}^{n+1} - 0) \cong \mathbb{C}^{n+1}$ define

$$\langle V, W \rangle_x = \frac{\mathrm{Re}(V,W)}{(Z,Z)}.$$

Notice that the metric $\langle \bullet, \bullet \rangle$ restricted to $S^{2n+1} \subseteq \mathbb{C}^{n+1} - 0$ coincides with the one induced from $\mathbb{R}^{2n+2}$.

(1) Show that for all $0 \le \theta \le 2\pi$, $e^{i\theta}\colon S^{2n+1} \to S^{2n+1}$ is an isometry. Thus it is possible to define a Riemannian metric on $\mathbb{CP}^n$ such that $f\colon S^{2n+1} \to \mathbb{CP}^n$ is a Riemannian submersion.

(2) Show that in this metric, the sectional curvature of $\mathbb{CP}^n$ is

$$K(\sigma) = 1 + 3\cos^2\varphi$$

where σ is generated by the orthonormal pair X, Y and $\cos\varphi = \left\langle \overline{X}, i\overline{Y} \right\rangle$ where $\overline{X}$ is the horizontal lift of X.

(1) Multiplication by $z \in \mathbb{C}$ has differential $dz\colon T_v\mathbb{C}^n \to T_{zv}\mathbb{C}^n$, after identifying both naturally with $\mathbb{C}^n$, given by multiplication by z. Thus multiplication by a $\mu = e^{i\theta}$ satisfies for $V, W \in T_zS^{2n+1}$,

$$\langle d\mu(V), d\mu(W)\rangle = \langle \mu V, \mu W\rangle = \mathrm{Re}(\mu V, \mu W) = \mathrm{Re}|\mu|^2(V, W) = \mathrm{Re}(V, W) = \langle V, W\rangle.$$

Thus μ is an isometry as desired.

In order for f to be a Riemannian submersion, the isomorphism

$$df_p\colon (T_pS^{2n+1})^h \xrightarrow{\sim} T_{fp}\mathbb{C}\mathrm{P}^n$$

must preserve the metric. This uniquely defines the metric on $T_{fp}\mathbb{C}\mathrm{P}^n$: define $\langle v, u\rangle_{\mathbb{C}\mathrm{P}^n} = \langle \overline{v}, \overline{u}\rangle_{S^{2n+1}}$ where $\overline{v}, \overline{u}$ are the unique horizontal lifts of v, u.

Now if $fp = fq$ then $q = \mu p$ for some $\mu = e^{i\theta}$ and because $f = f \circ \mu$,

$$df_q = df_p \circ d\mu_p$$

so that if $df_q(v') = df_p(v)$ then as μ is an isometry, $v' = d\mu_p(v)$ and since μ is an isometry this means the metric on $\mathbb{C}\mathrm{P}^n$ is well-defined.

This requires that μ be an isometry between the horizontal subspaces, so that $d\mu_p$ maps horizontal vectors to horizontal vectors. Because horizontal vectors are those in the kernel of df, and since $f = f \circ \mu$, if $df(v) = 0$ then $df \circ d\mu(v) = df(v) = 0$ as desired.

Note that the horizontal lift of a vector field $X \in \mathfrak{X}(M)$ exists independently of whether or not M's metric is smooth; as long as f is a surjective submersion which maps horizontal vectors isometrically. And the horizontal lift is smooth (when $\overline{M}$'s metric is); the proof given in the previous homework doesn't have an assumption on M's metric's smoothness. But then $g(X, Y) \circ f = \overline{g}(\overline{X}, \overline{Y})$ is smooth (since $\overline{g}, \overline{X}, \overline{Y}$ are) and since f is a surjective submersion, $g(X, Y)$ is smooth for all $X, Y \in \mathfrak{X}(M)$ so g is. That is to say, if there is a surjective submersion $f\colon \overline{M} \to M$ between manifolds equipped with metrics, where $\overline{M}$'s is smooth, such that f is an isometry on $\overline{M}$'s horizontal vectors (relative to f), then M's metric is smooth.

This is satisfied in our case, and so the metric defined on $\mathbb{C}\mathrm{P}^n$ is smooth. Thus this defines a smooth metric on $\mathbb{C}\mathrm{P}^n$ as desired.

(2) Let Z be the position vector describing S^{2n+1}, i.e. $Z_v = v$ for $v \in S^{2n+1}$. Because its extension to $\mathbb{C}^n$ is smooth, so $Z \in \Gamma(T\mathbb{C}^n|_{S^{2n+1}})$.

We know that S^n is the level set of 1 under $N\colon \mathbb{R}^{n+1} - 0 \to \mathbb{R}$, $N(x) = |x|^2$ which is a submersion. Thus T_pS^n consists of vectors $v \in T_p\mathbb{R}^{n+1}$ where $dN(v) = 0$. And the Jacobian of N at p is $2p$, so $dN(v) = 2(p, v)$, i.e. $T_pS^n = p^\perp \subseteq \mathbb{R}^{n+1}$.

In particular, $ip \in T_pS^{2n+1}$ since

$$\langle p, ip\rangle_p = \frac{\mathrm{Re}(p, ip)}{(p, p)} = \frac{-\mathrm{Re}\, i(p, p)}{(p, p)} = 0.$$

Thus iZ (which maps $p \mapsto ip$) is tangent to S^{2n+1}, so it defines a smooth vector field $iZ \in \mathfrak{X}(S^{2n+1})$.

Let $X, Y \in \mathfrak{X}(\mathbb{C}\mathrm{P}^n)$, and let $\alpha\colon (-\varepsilon, \varepsilon) \to S^{2n+1}$ be a smooth curve with $\alpha(0) = p$ and $\alpha'(0) = \overline{X}(p)$. Let $\overline{\nabla}$ be the Levi-Civita connection of S^{2n+1}, then

$$(\overline{\nabla}_{\overline{X}} iZ)_p = \left.\frac{d}{dt}\right|_{t=0} iZ \circ \alpha(t) = \left.\frac{d}{dt}\right|_{t=0} i\alpha(t) = i\alpha'(0) = i\overline{X}(p).$$

That is to say,

$$\overline{\nabla}_{\overline{X}} iZ = i\overline{X}.$$

Now,

$$\langle \overline{\nabla}_{\overline{X}}\overline{Y}, iZ\rangle = \overline{X}\langle \overline{Y}, iZ\rangle - \langle \overline{Y}, \overline{\nabla}_{\overline{X}} iZ\rangle = -\langle \overline{Y}, i\overline{X}\rangle,$$

where the last equality is due to $\overline{Y}$ being horizontal and iZ being vertical, as well as what we just showed. So we have

$$\langle [\overline{X}, \overline{Y}], iZ \rangle = \langle \overline{\nabla}_{\overline{X}} \overline{Y} - \overline{\nabla}_{\overline{Y}} \overline{X}, iZ \rangle = -\langle \overline{Y}, i\overline{X} \rangle + \langle \overline{X}, i\overline{Y} \rangle.$$

By definition we have

$$\begin{aligned} \langle v, iu \rangle_p &= \frac{\operatorname{Re}(v, iu)}{(p,p)} = \frac{-\operatorname{Re} i(v,u)}{(p,p)} = \frac{\operatorname{Im}(v,u)}{(p,p)}, \\ \langle iv, u \rangle_p &= -\frac{\operatorname{Im}(v,u)}{(p,p)}, \end{aligned}$$

so that $\langle iv, u \rangle_p = -\langle v, iu \rangle_p$. Thus

$$\langle [\overline{X}, \overline{Y}], iZ \rangle = 2\langle \overline{X}, i\overline{Y} \rangle = 2\cos\varphi.$$

By the previous question,

$$K(\sigma) = \overline{K}(\overline{\sigma}) + \frac{3}{4} \left| [\overline{X}, \overline{Y}]^v \right|^2.$$

Because the vertical tangent space has dimension 1, it is spanned by iZ, which has unit norm. Thus

$$[\overline{X}, \overline{Y}]^v = \langle [\overline{X}, \overline{Y}], iZ \rangle iZ = (2\cos\varphi) iZ$$

which has norm squared of $4\cos^2\varphi$. Because the unit sphere has sectional curvature of 1, we get that

$$K(\sigma) = 1 + 3\cos^2\varphi$$

as desired.

13.4 Problem

Let $U \subseteq \mathbb{R}^n$ be open and $f\colon U \to \mathbb{R}$ be smooth. Define

$$M = \{(\overline{x}, y) \mid \overline{x} \in U,\ y = f(\overline{x})\} \subseteq \mathbb{R}^{n+1}$$

and consider the coordinate chart $\mathbf{x}\colon U \to M$ by $\mathbf{x}(\overline{x}) = (\overline{x}, f(\overline{x}))$. Prove the following.

(1) A unit normal field to M is given by

$$\nu \circ \mathbf{x} = \frac{(-\nabla f, 1)}{\sqrt{1 + |\nabla f|^2}}.$$

(2) The second fundamental form of M is given by

$$\langle \mathrm{II}(\partial_i, \partial_j), \nu \rangle \circ \mathbf{x} = \frac{\operatorname{Hess} f(\partial_i, \partial_j)}{\sqrt{1 + |\nabla f|^2}}.$$

(3) The mean curvature of M is given by

$$\langle H, \nu \rangle \circ \mathbf{x} = \operatorname{div}\left(\frac{\nabla f}{\sqrt{1 + |\nabla f|^2}} \right).$$

(4) If $U = \mathbb{R}^n$ and f is linear, then M is a minimal surface.

(5) If $U = \{u \in \mathbb{R}^2 \mid |u| > a\}$ and $f = a\cosh^{-1}(|u|/a)$ then M is a minimal surface.

(1) Define $F : U \times \mathbb{R} \to \mathbb{R}$ by $F(\overline{x}, y) = y - f(\overline{x})$. Then $\nabla F = (-\nabla f, 1)$ so that F is a submersion and thus $M = F^{-1}0$ is embedded. And thus for $p \in M$, $T_pM = \ker \nabla F$ which are all the vectors orthogonal to $(-\nabla f, 1)$. Normalizing this, we see that a normal vector field is

$$\frac{(-\nabla f, 1)}{\sqrt{1 + |\nabla f|^2}}$$

as desired.

(2) Let us denote by $\widetilde{\partial}_i$ the coordinate frame induced by $(U, \mathbf{x})$, and ∂_i the Euclidean coordinate frame. We want to show

$$\left\langle \mathrm{II}(\widetilde{\partial}_i, \widetilde{\partial}_j), \nu \right\rangle \circ \mathbf{x} = \frac{\mathrm{Hess} f(\partial_i, \partial_j)}{\sqrt{1 + |\nabla f|^2}}.$$

Because $\mathrm{II}(X, Y) = \overline{\nabla}_X Y - \nabla_X Y$ where $\overline{\nabla}$ is the Euclidean connection and ∇ is M's, since $\nabla_X Y \in \mathfrak{X}(M)$ is orthogonal to ν,

$$\langle \mathrm{II}(X, Y), \nu \rangle = \langle \overline{\nabla}_X Y, \nu \rangle.$$

From exercise 8 we know

$$\widetilde{\partial}_i = f_i \partial_{n+1} + \partial_i, \qquad f_i = \frac{\partial f}{\partial \partial x_i}.$$

And thus

$$\left\langle \mathrm{II}(\widetilde{\partial}_i, \widetilde{\partial}_j), \nu \right\rangle = \left\langle \overline{\nabla}_{\widetilde{\partial}_i} \widetilde{\partial}_j, \nu \right\rangle = \langle \overline{\nabla}_{\partial_i + f_i \partial_{n+1}} (\partial_j + f_j \partial_{n+1}), \nu \rangle.$$

Because $\overline{\nabla}_{\partial_i} \partial_j = 0$ for all i, j, we see that this is equal to

$$= \langle \overline{\nabla}_{\partial_i} (f_j \partial_{n+1}) + f_i \overline{\nabla}_{\partial_{n+1}} (f_j \partial_{n+1}), \nu \rangle = \langle f_{ij} \partial_{n+1}, \nu \rangle.$$

Note the last equality is due to f not being a function of x_{n+1} so that $\partial_{n+1} f_j = 0$.

From the previous point, this is equal to

$$\left\langle \mathrm{II}(\widetilde{\partial}_i, \widetilde{\partial}_j), \nu \right\rangle = \frac{f_{ij}}{\sqrt{1 + |\nabla f|^2}}.$$

Now we compute $\mathrm{Hess} f(\partial_i, \partial_j)$ from definition:

$$\mathrm{Hess} f(\partial_i, \partial_j) = \langle \overline{\nabla}_{\partial_i} \nabla f, \partial_j \rangle = \left\langle \overline{\nabla}_{\partial_i} \sum_\ell f_\ell \partial_\ell, \partial_j \right\rangle = \sum_\ell \langle f_{i\ell} \partial_\ell, \partial_j \rangle = f_{ij}$$

which gives us the desired result.

(3) Firstly, mean curvature can be computed in the coordinate frame $\widetilde{\partial}_i$ by

$$H = g^{ij} \mathrm{II}(\widetilde{\partial}_i, \widetilde{\partial}_j),$$

and so by the previous point

$$\langle H, \nu \rangle = g^{ij} \left\langle \mathrm{II}(\widetilde{\partial}_i, \widetilde{\partial}_j), \nu \right\rangle = g^{ij} \frac{f_{ij}}{\sqrt{1 + |\nabla f|^2}}.$$

Recall that

$$g_{ij} = f_i f_j + \delta_{ij}$$

so that

$$g = I + (\nabla f)(\nabla f)^\top.$$

The ShermanMorrison formula then gives us g^{-1}:

$$g^{-1} = I - \frac{(\nabla f)(\nabla f)^\top}{1+|\nabla f|^2} \Longrightarrow g^{ij} = \delta_{ij} - \frac{f_i f_j}{1+|\nabla f|^2}.$$

Thus

$$\langle H, \nu\rangle = \left(\delta^{ij} - \frac{f_i f_j}{1+|\nabla f|^2}\right)\frac{f_{ij}}{\sqrt{1+|\nabla f|^2}} = \frac{f_{ii}}{\sqrt{1+|\nabla f|^2}} - \frac{f_i f_j f_{ij}}{(1+|\nabla f|^2)^{3/2}}$$

where the free indices are summed over.

On the other hand, because ∂_i forms a geodesic frame we know that

$$\operatorname{div}\left(\frac{\nabla f}{\sqrt{1+|\nabla f|^2}}\right) = \partial_i \frac{f_i}{\sqrt{1+|\nabla f|^2}} = \frac{f_{ii}}{\sqrt{1+|\nabla f|^2}} - \frac{f_i \partial_i (1+|\nabla f|^2)}{2(1+|\nabla f|^2)^{3/2}}.$$

A quick computation shows

$$\partial_i (1+|\nabla f|^2) = 2 f_j f_{ij}$$

and so

$$\operatorname{div}\left(\frac{\nabla f}{\sqrt{1+|\nabla f|^2}}\right) = \frac{f_{ii}}{\sqrt{1+|\nabla f|^2}} - \frac{f_i f_j f_{ij}}{(1+|\nabla f|^2)^{3/2}} = \langle H, \nu\rangle$$

as desired.

(4) In this point and the next, let

$$X = \frac{\nabla f}{\sqrt{1+|\nabla f|^2}}.$$

If f is linear, then ∇f is constant and so X is constant. Thus ∇X is zero, so its trace is zero, and thus $\operatorname{div} X = 0$ meaning $\langle H, \nu\rangle = 0$. Since H is orthogonal to M, this means $H = 0$ so M is minimal.

(5) Let us write (x, y) instead of (u_1, u_2) and let $r = \sqrt{x^2+y^2}$. Then our function is $f(r) = a\cosh^{-1}(r/a)$ whose derivative is

$$f'(r) = \frac{1}{\sqrt{r^2/a^2 - 1}} = \frac{a}{\sqrt{r^2 - a^2}}$$

Because $\nabla r = (x/r, y/r)$ we get that

$$\nabla f = \frac{a}{\sqrt{r^2-a^2}}\left(\frac{x}{r}, \frac{y}{r}\right) \Longrightarrow |\nabla f|^2 = \frac{a^2}{r^2-a^2}.$$

And so

$$\sqrt{1+|\nabla f|^2} = \frac{r}{\sqrt{r^2-a^2}},$$

thus

$$X = \frac{a}{r}\left(\frac{x}{r}, \frac{y}{r}\right) = \frac{a}{r^2}(x, y).$$

Its divergence is then

$$\frac{1}{a}\cdot \operatorname{div} X = \operatorname{div}\left(\frac{x}{r^2}, \frac{y}{r^2}\right) = \frac{\partial}{\partial x}\frac{x}{r^2} + \frac{\partial}{\partial y}\frac{y}{r^2}.$$

Computing shows that

$$\frac{\partial}{\partial x}\frac{x}{r^2} = \frac{\partial}{\partial x}\frac{x}{x^2+y^2} = \frac{r^2-2x^2}{r^4}$$

so that

$$\frac{1}{a}\cdot \mathrm{div} X = \frac{2}{r^2} - 2\frac{x^2+y^2}{r^4} = 0,$$

meaning $\langle H, \nu\rangle = \mathrm{div} X = 0$ and thus $H = 0$ so M is minimal.